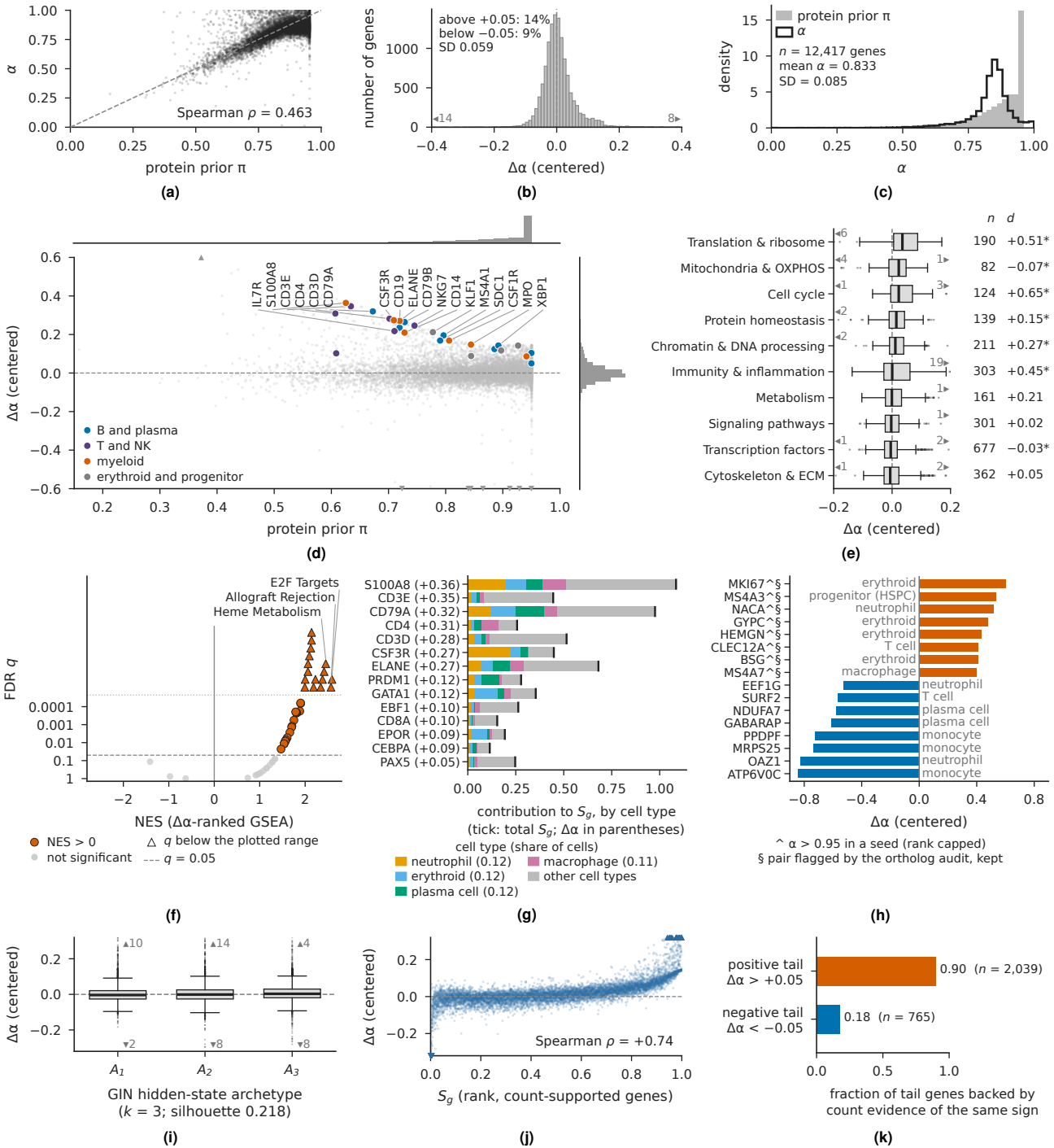

Spearman ρ = 0.463
protein prior π
α
(a)
above +0.05: 14%
below −0.05: 9%
SD 0.059
number of genes
Δα (centered)
(b)
protein prior π
α
n = 12,417 genes
mean α = 0.833
SD = 0.085
density
(c)
IL7R
S100A8
CD3E
CD4
CD3D
CD79A
CSF3R
CD19
ELANE
CD79B
NKG7
CD14
KLF1
MS4A1
SDC1
CSF1R
MPO
XBP1
B and plasma
T and NK
myeloid
erythroid and progenitor
Δα (centered)
protein prior π
(d)
n
d
Translation & ribosome 190 +0.51*
Mitochondria & OXPHOS 82 −0.07*
Cell cycle 124 +0.65*
Protein homeostasis 139 +0.15*
Chromatin & DNA processing 211 +0.27*
Immunity & inflammation 303 +0.45*
Metabolism 161 +0.21
Signaling pathways 301 +0.02
Transcription factors 677 −0.03*
Cytoskeleton & ECM 362 +0.05
Δα (centered)
(e)
E2F Targets
Allograft Rejection
Heme Metabolism
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
S100A8 (+0.36)
CD3E (+0.35)
CD79A (+0.32)
CD4 (+0.31)
CD3D (+0.28)
CSF3R (+0.27)
ELANE (+0.27)
PRDM1 (+0.12)
GATA1 (+0.12)
EBF1 (+0.10)
CD8A (+0.10)
EPOR (+0.09)
CEBPA (+0.09)
PAX5 (+0.05)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
neutrophil (0.12)
macrophage (0.11)
erythroid (0.12)
other cell types
plasma cell (0.12)
(g)
MKI67^§ erythroid
MS4A3^§ progenitor (HSPC)
NACA^§ neutrophil
GYPC^§ erythroid
HEMGN^§ erythroid
CLEC12A^§ T cell
BSG^§ erythroid
MS4A7^§ macrophage
EEF1G neutrophil
SURF2 T cell
NDUFA7 plasma cell
GABARAP plasma cell
PPDPF monocyte
MRPS25 monocyte
OAZ1 neutrophil
ATP6V0C monocyte
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.218)
(i)
Spearman ρ = +0.74
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.90 (n = 2,039)
negative tail
Δα < −0.05
0.18 (n = 765)
fraction of tail genes backed by
count evidence of the same sign
(k)